

Mathematical Problem-1

CSE4261: Neural Network and Deep Learning
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1. Assume a CNN-based 3-class classifier which can be used for CAM.
Let a 5x5 image be passed through the assumed CNN-based 3-class classifier.

$$I = \begin{bmatrix} 1 & 2 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 3 & 4 & 3 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

Let the last convolutional layer of a CNN has three feature maps (size 2x2)

$$f_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad f_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad f_3 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Let the fully connected layer weights (for class c)

$$w^c = [2, -1, 1]$$

- a. Compute Global Average Pooling (GAP) outputs F_1, F_2, F_3 .
- b. Compute the class score:

$$S_c = \sum_k w_k^c F_k$$

- c. Compute the Class Activation Map:

$$M_c(x, y) = \sum_k w_k^c f_k(x, y)$$

- d. Normalize the CAM:

$$M' = \frac{M - \min(M)}{\max(M) - \min(M)}$$

- e. Explain how this 2x2 CAM would be mapped back to the 5x5 input image.
2. Show steps of CAM for your own three sets of $\{I, f_1, f_2, f_3, w^c\}$.
 3. Show steps of Grad-CAM for the above network for the above image.
 4. Show Grad-CAM when multiple FC layers exist.
 5. Visualize before vs after ReLU as heatmaps
 6. Show what happens if we DON'T use ReLU (full attribution).
 7. Say model $F(x_1, x_2) = x_1^2 + 2x_2$, input $x = (2, 3)$, baseline $x' = (0, 0)$, $m = 10$. compute the Integrated Gradients (IG) using a discrete approximation. Show all steps as in a real implementation of the IG algorithm.
 8. Show all steps as in a real implementation of the IG algorithm for your own three sets of $\{F, x, x', m\}$.
 9. Compare Grad-CAM vs Integrated Gradients (IG) using the same situation.